

D687

Computer Science Project Development with a Team

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

Your team built a model that predicts equipment failures with 0.91 AUC using gradient boosting on three years of sensor data. The team met every Thursday during the build. You are writing the section for the chief operating officer. Which sentence belongs there?

- A. "The model flags likely failures about a week early, which could cut unplanned downtime costs by an estimated \$400,000 a year."
- B. "The model will prevent all equipment failures."
- C. "The gradient-boosted model reached 0.91 AUC on held-out sensor data."
- D. "The team met every Thursday to review progress."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which item belongs in the technical report rather than the executive summary?

- A. The funding request.
- B. Nothing; both documents should be identical.
- C. Projected annual savings.
- D. The hyperparameters and the validation method.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An executive summary draft opens with three paragraphs on the history of the team's coding choices and states the funding request on page 4. The full proposal is 22 pages. The CIO reads on a phone between meetings. What is the best revision?

- A. Open with the problem, the solution, the expected value, and the specific funding request on the first page.
- B. Add more coding history so the CIO sees the effort.
- C. Remove the funding request so the summary seems less pushy.
- D. Shorten the full proposal to 10 pages and leave the summary as is.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A project costs \$80,000 to build and \$20,000 a year to run. It saves \$70,000 a year. A team of four built it in 10 weeks. What is its ROI over three years?

- A. About 33%
- B. 150%
- C. 50%
- D. 162.5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Stakeholders require at least 25% ROI over two years. Total two-year costs are \$120,000. What minimum total benefit is needed?

- A. \$96,000
- B. \$160,000
- C. \$150,000
- D. \$30,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Requirements for an appointment app at a clinic with three locations include the items below. Which is a non-functional requirement?

- A. The project must finish by March.
- B. Pages must load within 2 seconds for 95% of requests.
- C. Patients can cancel an appointment online.
- D. The system sends a reminder text 24 hours before each visit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Which requirement is functional?

- A. All stored data must be encrypted.
- B. Managers can export the monthly revenue report as a CSV file.
- C. The interface must be easy to learn.
- D. The system must be available 99.9% of the time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

You are peer-reviewing an 18-page technical report written by the team's most experienced coder. Which comment is most useful?

- A. "You always write unclearly."
- B. "You're the best coder here, so I'll skip my review."
- C. "Great job, looks good!"
- D. "Section 3 reports 94% accuracy but doesn't say which data it was tested on; adding the test-set size and date range would let readers trust it."

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Three peer reviewers say your proposal's cost section is confusing; one says the introduction is too long. You spent two late nights on the cost section and like it. The deadline is five days away. What is the best response?

- A. Explain to reviewers why the cost section is already clear.
- B. Trim the introduction first because it is quickest.
- C. Ask reviewers what confused them, revise the cost section first, then consider trimming the introduction.
- D. Rewrite the entire proposal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A team's technical report states, "The model works well." It leaves out the data's date range, the preprocessing steps, and a known weakness on rural stores. The report includes 12 colorful charts. What most needs to be added?

- A. A methods section (data, preprocessing, validation) and a limitations section that names the rural-store weakness.
- B. More charts.
- C. A statement that the model is 100% reliable.
- D. Leave out the rural weakness to avoid alarming leaders.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A project costs \$90,000 up front. Net savings are \$30,000 in year 1 and \$40,000 in each of years 2 and 3. The pitch deck has 15 slides. What is the payback period?

- A.** 3 years
- B.** 2.5 years
- C.** About 0.44 years
- D.** 2.25 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Leadership wants a two-year payback on a \$150,000 project with equal yearly savings. What yearly savings are needed?

- A.** \$50,000
- B.** \$150,000
- C.** \$300,000
- D.** \$75,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

Your team built a model that predicts equipment failures with 0.91 AUC using gradient boosting on three years of sensor data. ~~The team met every Thursday during the build.~~ You are writing the section for the chief operating officer. Which sentence belongs there?

Target word: executive-appropriate content **Decides:** The reader is an executive.

A	Correct	Speaks in outcomes and dollars.
B	Absolute language	Overclaims; no model prevents all failures.
C	Adjacent category	Accurate but aimed at a technical reader.
D	Camouflage	Process trivia, not value.

2. Answer D Concept 1 • transfer

Which item belongs in the technical report rather than the executive summary?

Target word: technical content — reversed **Decides:** Implementation detail for engineers.

A	Confusion set	The ask belongs up front in the executive summary.
B	Absolute language	Different audiences need different content.
C	Inverted operation	A headline business figure for executives.
D	Correct	Needed to reproduce and assess the model.

3. Answer A Concept 2

An executive summary draft opens with three paragraphs on the history of the team's coding choices and states the funding request on page 4. ~~The full proposal is 22 pages.~~ The CIO reads on a phone between meetings. What is the best revision?

Target word: fix the summary **Decides:** The ask and value are buried.

A	Correct	Busy executives decide from the first page.
B	Inverted operation	Buries the ask deeper.
C	Dropped element	Drops the decision the summary exists to get.
D	Camouflage	The page count does not fix the summary's order.

4. Answer C Concept 3

A project costs \$80,000 to build and \$20,000 a year to run. It saves \$70,000 a year. ~~A team of four built it in 10 weeks.~~ What is its ROI over three years?

Target word: three-year ROI **Decides:** Costs = 80,000 + 3 × 20,000 = 140,000; benefits = 210,000.

A	Confusion set	70,000 / 210,000 divides by benefits.
B	Inverted operation	210,000 / 140,000 is the benefit-cost ratio, not net return.
C	Correct	$(210,000 - 140,000) / 140,000 = 0.50$.
D	Dropped element	$(210,000 - 80,000) / 80,000$ leaves out run costs.

5. Answer C Concept 3 · transfer

Stakeholders require at least 25% ROI over two years. Total two-year costs are \$120,000. What minimum total benefit is needed?

Target word: solve for benefit **Decides:** Benefit = cost $\times$ 1.25.

A	Inverted operation	$120,000 \times 0.80$ goes the wrong direction.
B	Confusion set	$120,000 / 0.75$ sets a margin on benefit, not a return on cost.
C	Correct	$120,000 \times 1.25 = 150,000$.
D	Dropped element	The required net gain, not total benefit.

6. Answer B Concept 4

Requirements for an appointment app at a clinic ~~with three locations~~ include the items below. Which is a non-functional requirement?

Target word: identify non-functional **Decides:** Non-functional = how well, not what.

A	Adjacent category	A project constraint, not a system requirement.
B	Correct	A performance quality attribute.
C	Inverted operation	A function the system performs.
D	Confusion set	Also a function.

7. Answer B Concept 4 · transfer

Which requirement is functional?

Target word: identify functional — reversed **Decides:** Describes a behavior.

A	Confusion set	A security quality, non-functional.
B	Correct	States something the system does.
C	Adjacent category	Usability is non-functional.
D	Inverted operation	Availability is non-functional.

8. Answer D Concept 5

You are peer-reviewing an ~~18-page~~ technical report ~~written by the team's most experienced coder~~. Which comment is most useful?

Target word: useful feedback **Decides:** Specific, work-focused, actionable.

A	Absolute language	Personal and sweeping.
B	Camouflage	Experience does not replace review.
C	Dropped element	No specifics to act on.
D	Correct	Specific, tied to the work, and offers a fix.

9. Answer C Concept 6

Three peer reviewers say your proposal's cost section is confusing; one says the introduction is too long. ~~You spent two late nights on the cost section and like it.~~ The deadline is five days away. What is the best response?

Target word: integrating feedback **Decides:** Three of four reviewers flag the same issue.

A	Inverted operation	Defends instead of listening.
B	Dropped element	Drops the frequency and impact of the cost feedback.
C	Correct	Repeated, high-impact feedback takes priority.
D	Absolute language	Overreacts; only specific sections were flagged.

10. Answer A Concept 7

A team's technical report states, "The model works well." It leaves out the data's date range, the preprocessing steps, and a known weakness on rural stores. ~~The report includes 12 colorful charts.~~ What most needs to be added?

Target word: missing report content **Decides:** No methods and no limitations.

A	Correct	Lets readers reproduce and judge the results.
B	Camouflage	Visuals do not replace methods.
C	Absolute language	Unsupported and false.
D	Inverted operation	Hiding limits undermines trust.

11. Answer B Concept 8

A project costs \$90,000 up front. Net savings are \$30,000 in year 1 and \$40,000 in each of years 2 and 3. ~~The pitch deck has 15 slides.~~ What is the payback period?

Target word: payback period **Decides:** Cumulative: 30,000 after year 1; 70,000 after year 2.

A	Dropped element	90,000 / 30,000 uses only year 1's savings.
B	Correct	\$20,000 still owed after year 2; 20,000 / 40,000 = 0.5 of year 3.
C	Inverted operation	40,000 / 90,000 divides the wrong way.
D	Wrong stem number	90,000 / 40,000 ignores the lower first year.

12. Answer D Concept 8 • transfer

Leadership wants a two-year payback on a \$150,000 project with equal yearly savings. What yearly savings are needed?

Target word: solve for savings **Decides:** Savings $\times 2 = 150,000$.

A	Wrong stem number	Divides by 3 years instead of 2.
B	Dropped element	Drops the two-year spread.
C	Inverted operation	Multiplies by 2 instead of dividing.
D	Correct	$150,000 / 2 = 75,000$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Write for the audience	Executives need business outcomes, cost, and risk; technical readers need methods, metrics, and implementation detail.	Giving executives model metrics, or giving engineers only savings claims.	“Which sentence belongs in the executive section?”
2	Executive summary structure	Lead with the problem, the proposed solution, the expected value, and the specific ask, on the first page.	Opening with background and burying the request.	A summary whose funding request appears pages later.
3	Return on investment	$ROI = (\text{total benefits} - \text{total costs}) / \text{total costs}$ over the same period, including ongoing costs.	Leaving out run costs, or reporting benefit/cost as ROI.	A build cost, a yearly run cost, and yearly savings.
4	Functional vs. non-functional requirements	Functional requirements say what the system does; non-functional requirements say how well (speed, availability, security, usability).	Calling a feature a quality attribute or vice versa.	A list of “shall” statements and “which is non-functional?”
5	Giving useful peer feedback	Good feedback is specific, tied to the work (not the person), and suggests an improvement.	Vague praise or sweeping personal judgments.	“Which review comment is most useful?”
6	Receiving and integrating feedback	Listen, clarify, look for themes across reviewers, and act first on repeated, high-impact issues.	Defending the work, or fixing the easiest comment first.	Several reviewers raising the same point.
7	Technical report: methods and limits	A technical report must let a reader reproduce and judge results: data, preprocessing, validation, results, and known limitations.	Adding polish while omitting methods, or hiding weaknesses.	A report that says “it works well” without evidence.
8	Payback period	Payback = time until cumulative net savings equal the up-front cost, interpolating within the final year.	Dividing cost by one year’s savings when savings vary.	An up-front cost and year-by-year savings.

Course scope (public description): Computer Science Project Development with a Team has students prepare a prior project for submission to a mock technical and executive leadership team. This course expands on the coding work done in a previous course, asking students to submit three artifacts. The final artifact is a business proposal aimed at convincing stakeholders to implement the project, which includes an executive summary of product requirements directed at the IT audience, as well as a technical report of the fully functional data product intended to solve a real-world problem. Artifacts are evaluated by peer team members prior to submission, and students practice giving, receiving, and integrating feedback into their work process.